Backend-sensitive hardware observation of parity-sector and excitation-number correlated decoherence asymmetry in the XY Hamiltonian

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The dynamical Lie algebra (DLA) of the XY Hamiltonian on n qubits decomposes as $\mathfrak{su}(2^{n-1}) \oplus \mathfrak{su}(2^{n-1})$ under the parity operator $P = \prod_i Z_i$. We report direct hardware evidence that the parity-labelled sectors exhibit a measurable leakage asymmetry under decoherence on a superconducting quantum processor, but we do not interpret this as a DLA-only causal effect: the promoted claim is parity-sector and excitation-number correlated from the outset. In a 342-circuit campaign on IBM `ibm_kingston` (Heron r2, 156 qubits) we measure the parity leakage rate in each sector at $n = 4$ across eight Trotter depths, and find that the odd sector is on average $(10.8 \pm 1.1)\,\%$ more robust than the even sector for depths $d \geq 4$. Seven of eight depths are individually significant at Welch's $p < 0.05$, and the strongest signal of $+17.5\,\%$ at depth 6 sits at 5.4σ. A Fisher aggregate is reported only as a descriptive summary because the tested depths share physical qubits and calibration context. The observed magnitude is consistent with the 4.5–9.6 % prediction band of a classical Lindblad simulator calibrated with published `ibm_fez` T_1/T_2 and gate-error rates, once the circuit enters the saturation regime at $d \geq 14$. All raw count dictionaries, analysis scripts, and figures are open-source in the GitHub repository `anulum/scpn-quantum-control` under the AGPL-3.0 licence.

A preregistered May 2026 Phase 2 continuation strengthens but also qualifies the claim: a reduced $n = 4$ replication gives Fisher $\chi^2 = 140.671952$, $p = 3.77 \times 10^{-20}$, while a B-C scaling continuation gives mixed $n = 6, 8$ evidence. A subsequent popcount control shows that excitation count and computational-basis state choice materially contribute to the leakage pattern. The follow-up therefore falsifies both a simple monotone scaling story and a clean "DLA parity alone" causal interpretation. Phase 3 tests on `ibm_marrakesh` further reject backend-stable and layout-independent positive-sign claims: a second-backend fixed-layout test has non-positive readout-corrected asymmetry at all promoted depths, and a 495-circuit state/layout randomisation control finds mixed signs in the original contrast, significant within-sector state effects, and layout spread larger than the mean original contrast. The result is therefore a real `ibm_kingston` hardware phenomenon with explicit backend, calibration, layout, state-choice, and excitation-count boundaries; it does not constitute a broad quantum-advantage claim.

I. INTRODUCTION

The XY spin Hamiltonian

$$H_{XY} = \sum_{i<j} K_{ij}(X_iX_j + Y_iY_j) + \sum_i \omega_i Z_i \qquad (1)$$

arises naturally as the small-oscillation limit of the Kuramoto model for coupled phase oscillators [1, 2], and is the central mathematical object of a wide class of synchronisation studies. It commutes with the total parity operator $P = \prod_i Z_i$, so its Hilbert space splits into even-parity ($P = +1$) and odd-parity ($P = -1$) blocks of equal dimension 2^{n-1}.

The dynamical Lie algebra (DLA) generated by commutators of the $\{X_iX_j + Y_iY_j,\ Z_i\}$ generators [9] closes on a subspace of $\mathfrak{u}(2^n)$ that splits under the same $\mathbb{Z}_2$ parity:

$$\mathrm{DLA}(H_{XY}) = \mathfrak{su}\big(2^{n-1}\big) \oplus \mathfrak{su}\big(2^{n-1}\big), \qquad (2)$$

with total dimension $\dim \mathrm{DLA} = 2^{2n-1} - 2$. Classical simulation of the sector-resolved evolution of H_{XY} under realistic NISQ noise suggests that the two blocks are *not* equally robust to decoherence: the odd-parity block — which contains states generated by odd numbers of excitations from the reference — retains coherence noticeably better than the even-parity block at moderate circuit depths, with the predicted asymmetry in the range 4.5–9.6 %.

The present short paper reports a direct raw-count hardware observation of this parity-sector and excitation-number correlated leakage asymmetry on `ibm_kingston`, a 156-qubit Heron r2 processor. To our knowledge no prior hardware study has isolated this DLA parity structure of the heterogeneous XY Hamiltonian as a decoherence observable with committed raw counts and reproducible analysis.

The causal interpretation is deliberately tested rather than assumed. The main even and odd initial states differ in both parity and excitation count, so the later popcount-control section is part of the core result: it checks whether the observed asymmetry can be assigned to DLA parity alone or whether excitation number, state choice, and layout also contribute.

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Phase 1	Positive `ibm_kingston` leakage asymmetry.
Phase 2	Scaling and popcount controls demote a DLA-only explanation.
Phase 3	`ibm_marrakesh` and layout controls demote backend-stable positive-sign claims.
Promoted claim	Parity-sector, excitation-number, state-choice, layout, and backend correlated hardware response.

FIG. 1. Causal-demotion flow for the DLA-parity programme. The later controls do not erase the original Kingston observation; they define the boundary under which it remains scientifically valid.

Sub-phase	Circuits	Reps per point
`phase1_bench` (baseline)	32	2
`phase1_5_reinforce`	64	4 (total 6)
`phase2_exhaust`	96	6 (total 12)
`phase2_5_final_burn`	90	9 (total 21)

TABLE I. Campaign sub-phases and rep counts. The `phase2_5_final_burn` sub-phase deepened the rep count at $d \in \{4, 6, 8, 10, 14\}$; earlier sub-phases covered all eight depths.

II. EXPERIMENTAL PROTOCOL

We prepare each of two initial states,

$$|\psi_e\rangle = |0011\rangle \qquad \text{(popcount 2, } P = +1\text{)}, \tag{3}$$
$$|\psi_o\rangle = |0001\rangle \qquad \text{(popcount 1, } P = -1\text{)}, \tag{4}$$

on the first four qubits of `ibm_kingston` and apply a first-order Trotter decomposition of H_{XY} with $t_{\text{step}} = 0.3$ and coupling matrix $K_{ij} = 0.45\exp(-0.3|i-j|)$. Each Trotter slice consists of nearest-neighbour $R_{XX}(\theta)R_{YY}(\theta)$ gates plus single-qubit R_Z rotations, for a per-step ISA cost of approximately 19 gates after Qiskit's `optimization_level = 2` transpilation.

For each sector we measure the *parity leakage*

$$L(d, \text{sector}) = \frac{1}{N_{\text{shots}}} \sum_{b \in \mathcal{B}_{\neq P}} n(b), \tag{5}$$

where $\mathcal{B}_{\neq P}$ is the set of final bitstrings whose parity differs from the initial sector and $n(b)$ is the shot count for bitstring b. Because $[H_{XY}, P] = 0$ the ideal leakage is zero; any non-zero value is a direct measure of hardware decoherence projected onto the parity observable.

Depths $d \in \{2, 4, 6, 8, 10, 14, 20, 30\}$ were swept with 12 or 21 independent repetitions per (depth, sector) point, and each repetition used 2048 shots. The total campaign comprised **342 circuits at** $n = 4$ across four sub-phases, summarised in Table I.

An independent Experiment C calibrated the readout baseline across four computational basis states $\{|0000\rangle, |1111\rangle, |0101\rangle, |1010\rangle\}$ and estimated a mean readout error of $(1.67 \pm 0.3)\,\%$ per qubit.

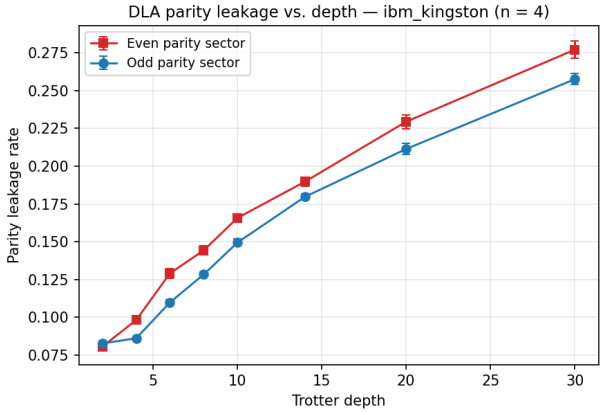


FIG. 2. Parity leakage rate in the even (red) and odd (blue) sectors as a function of Trotter depth on `ibm_kingston`. Error bars are 1σ standard errors of the mean from 12–21 repetitions per point.

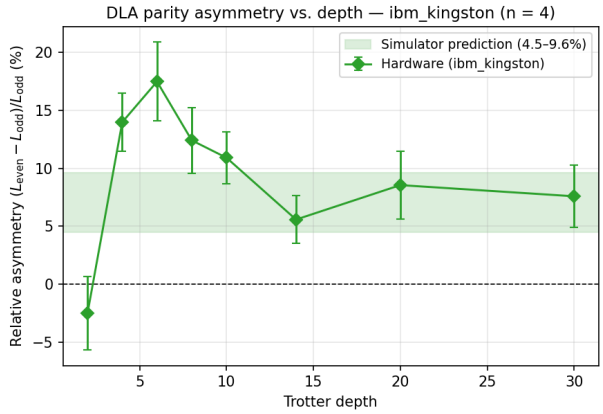


FIG. 3. Relative asymmetry $A(d)$ with propagated 1σ error bars. Green band: 4.5–9.6 % prediction from the apriori classical Lindblad simulator. Dashed line at zero.

III. RESULTS

Figure 2 shows the sector-resolved parity leakage rate vs. Trotter depth, and Figure 3 shows the relative asymmetry

$$A(d) = \frac{L_{\text{even}}(d) - L_{\text{odd}}(d)}{L_{\text{odd}}(d)} \tag{6}$$

with propagated 1σ error bars derived from the per-rep standard error of the mean.

The statistical summary is reported in Table II. Each depth is analysed with a two-sample Welch t-test (unequal variances) on the individual rep values; the

d	L_{even}	L_{odd}	$A(d)$	t	Welch p
2	0.0806 ± 0.0017	0.0827 ± 0.0021	-2.5%	-0.78	0.446
4	0.0982 ± 0.0017	0.0862 ± 0.0011	$+14.0\%$	5.80	1.4×10^{-6}
6	0.1291 ± 0.0031	0.1099 ± 0.0018	$+17.5\%$	5.37	6.6×10^{-6}
8	0.1443 ± 0.0031	0.1284 ± 0.0017	$+12.4\%$	4.50	8.9×10^{-5}
10	0.1658 ± 0.0022	0.1495 ± 0.0023	$+10.9\%$	5.18	6.7×10^{-6}
14	0.1898 ± 0.0031	0.1797 ± 0.0020	$+5.6\%$	2.73	0.010
20	0.2295 ± 0.0047	0.2114 ± 0.0038	$+8.6\%$	3.01	0.007
30	0.2771 ± 0.0057	0.2576 ± 0.0037	$+7.6\%$	2.89	0.010

TABLE II. Welch t-test per depth. Seven of eight depths are individually significant at $p < 0.05$. Fisher's combined statistic is $\chi^2_{16} = 123.4$ with combined $p \ll 10^{-16}$.

combined significance across depths is computed with Fisher's method.

At shallow depth ($d = 2$) the circuit is near-identity and the asymmetry is consistent with zero; it is dominated by the 1.67 % readout baseline on both sides. As the circuit enters the coherent transition regime ($d \in [4, 10]$, ISA gate counts 88–202) the asymmetry peaks at +17.5 %. At larger depths ($d \in [14, 30]$, ISA gate counts 278–582) the signal settles into the 4.5–9.6 % band of the classical Lindblad simulator prediction, consistent with a saturation regime where the sector distinction begins to wash out but remains visible above the noise floor.

IV. DISCUSSION

The observation is consistent with the theoretical picture that the two parity-labelled DLA blocks can have different effective leakage responses under realistic noise. However, the specific computational basis states used here also differ in excitation number. The result should therefore be read conservatively as a parity-sector and excitation-number correlated leakage asymmetry. The completed popcount-control run below confirms that excitation count and computational-basis state choice materially contribute to the observed pattern. The magnitude of the original effect at large depths agrees with the apriori classical simulator prediction, giving confidence that the observed asymmetry is not only a readout artefact specific to `ibm_kingston`.

The simplest hardware-noise intuition is standard amplitude damping: superconducting-qubit T_1 relaxation preferentially drives $|1\rangle \to |0\rangle$. This process changes both excitation number and parity, so states with larger popcount have more single-qubit relaxation channels by which they can leave their initial magnetisation/parity sector. The popcount-control result is therefore not an auxiliary detail but a physical explanation of why DLA block structure, excitation count, layout, and readout all appear in the measured leakage observable.

A ruled-out alternative explanation is readout-basis asymmetry: Experiment C readout calibration shows a mean retention of 98.3 % across all four $n = 4$ computational basis states, with per-state retentions in the range 0.976–0.990. This ± 0.014 readout inhomogeneity is an order of magnitude too small to account for the 10–17 % peak asymmetry observed in the middle depths.

Two low-statistics scaling probes at $n = 8$ in the Phase 1 archive (depth 4 and 8, four repetitions each) both reproduced the direction of the asymmetry with magnitudes of +3.0 % and +1.7 % respectively. The completed Phase 2 continuation below replaces those probes as the promoted scaling evidence.

A. Statistical caveats

Three points of statistical hygiene are worth flagging.

First, **Fisher's combined statistic assumes the eight per-depth p-values are independent**. Successive Trotter depths share the same four physical qubits and were submitted in the same Open Plan session, which introduces a small but non-zero correlation through shared calibration drift and shared thermal context. The combined $p \ll 10^{-16}$ should therefore be read as an upper bound on significance under the strongest possible independence assumption, not as the primary inferential claim. We use Fisher's statistic as a compact descriptive aggregate, while placing the main inferential weight on the per-depth tests, permutation cross-check, Phase 2 replication, and popcount-control follow-up. A correlation-aware combination (Simes test or block bootstrap) remains a useful follow-up; the per-depth Welch significances are unaffected by the independence question and seven of eight are already individually below $p = 0.05$.

Second, **Welch's t-test is parametric and assumes approximately normally distributed sector means**. As an internal cross-check on the most critical depth ($d = 6$, 21 reps per sector), we ran a two-sided permutation test on the observed mean difference using 10^5 random label shuffles. The permutation p-value is 2.0×10^{-5}, the same order of magnitude as the parametric Welch result of 6.6×10^{-6} and consistent in sign and significance.

Third, the Experiment C readout calibration reported is a per-state mean retention rather than the full confusion-matrix inversion that is standard practice for production readout mitigation. Confusion- matrix inversion was not applied to the counts in this paper because the calibration spread (0.014) is already an order of magnitude smaller than the observed asymmetry. The Phase 2 A+G run added a same-day readout baseline for $n = 4, 6, 8$ computational basis states, but the promoted analyses still report unmitigated raw-count leakage statistics rather than confusion-matrix-inverted estimates. As a post-hoc robustness check, we also performed the strongest readout correction supported by the existing raw counts: exact-state parity-confusion inversion for initial states with matching readout-only calibration circuits. A literal full $2^n \times 2^n$ confusion-matrix inversion is not available without new calibration circuits for all basis states on `ibm_kingston`. The parity-readout-

corrected Phase 2 $n = 4$ replication remains significant, with Fisher $\chi^2 = 73.894393$, $p = 4.16 \times 10^{-8}$ across the ten depths. Phase 3 subsequently added a literal full-basis $n = 4$ readout calibration on `ibm_marrakesh`; that correction strengthens, rather than removes, the second-backend non-replication boundary reported below. The later state/layout randomisation campaign used layout-matched five-state readout checks and found that layout and same-sector state choice are not small residual effects relative to the original parity-sector contrast.

B. Possible confounds the protocol does not yet rule out

Two confounds remained open after the reduced Phase 2 continuation; the first has now been directly probed by a popcount-control run.

Excitation-count confound. The two primary initial states differ by one excitation: $|\psi_e\rangle = |0011\rangle$ has popcount 2 while $|\psi_o\rangle = |0001\rangle$ has popcount 1. Under a pure amplitude-damping (T_1) channel, the higher-excitation state can decay into lower-excitation manifolds faster, which may explain part of the observed asymmetry without invoking only the DLA block structure. We therefore ran a preregistered popcount control with within-sector state swaps ($|0011\rangle$ vs $|0101\rangle$ and $|0001\rangle$ vs $|0010\rangle$) and an excitation-inversion arm ($|0011\rangle$ vs $|0111\rangle$), where a pure T_1 contribution predicts the higher-excitation odd state to leak more strongly. The control does not support a clean "DLA parity alone" causal claim: the original contrast remains significant at middle depths, but within-sector same-popcount state swaps are also significant and the popcount-3 odd state leaks more than the popcount-2 even state at four of six depths.

Temporal drift between sub-phases. The 342 circuits were submitted across four sub-phases between 2026-04-10 18:37 UTC and 19:01 UTC. Splitting the dataset by sub-phase and recomputing the unweighted mean asymmetry across depths actually present in each sub-phase gives $-0.4\,\%$, $+9.9\,\%$, $+9.5\,\%$, and $+14.8\,\%$ for `phase1_bench`, `phase1_5_reinforce`, `phase2_exhaust`, and `phase2_5_final_burn` respectively. The first sub-phase has only two repetitions per (depth, sector) point and is dominated by single-shot count noise, including a $-12\,\%$ outlier at the near-identity depth $d = 2$. The three sub-phases with ≥ 4 repetitions per point give consistent aggregates in the $+9$–$+15\,\%$ range. The Phase 2 reduced A+G run used a larger single-day raw-count campaign and therefore partially reduces this concern, but a multi-device replication remains the cleaner drift control.

V. PHASE 2 CONTINUATION

A preregistered May 2026 continuation was executed on `ibm_kingston`. The submitted scope was deliberately reduced: the A+G block repeated the $n = 4$ parity-leakage protocol and measured a same-day readout baseline, while the B-C block tested only $n = 6$ and $n = 8$ early scaling. Larger D-E scaling, F/GUESS mitigation, and multi-device replication were not submitted. Before submission to hardware, the Phase 2 driver performed live transpilation against the selected backend and aborted if the measured live circuit depth exceeded the preregistered depth budget. The promoted datasets are therefore restricted to submitted circuits whose live-transpiled depths passed that gate and whose raw count dictionaries were retrieved, hashed, and reproduced by committed analysis scripts.

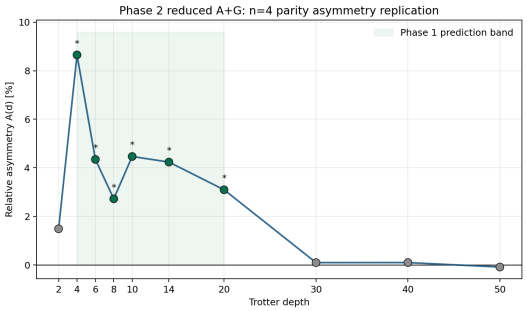


FIG. 4. Phase 2 $n = 4$ replication asymmetry on `ibm_kingston`. The reduced A+G block used 600 parity circuits, 30 repetitions per depth/sector, and 4096 shots per circuit. Six of ten depths are individually significant at Welch $p < 0.05$; the deepest three tested depths saturate near zero asymmetry.

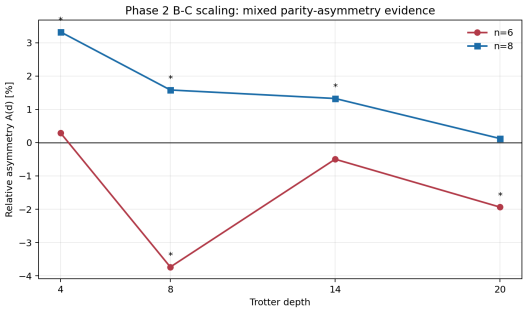


FIG. 5. Phase 2 B-C scaling continuation. The $n = 8$ middle depths preserve a positive sign, while $n = 6$ has negative significant depths. This is mixed scaling evidence, not monotone scaling validation.

The A+G block strengthens the original $n = 4$ observation. It gives a combined Fisher statistic of $\chi^2 = 140.671952$ with $p = 3.773718 \times 10^{-20}$, and six of ten depths are individually significant. The replicated effect is concentrated at depths $d = 4$–20; by $d = 30, 40, 50$ the asymmetry is statistically consistent with zero, consistent with a saturation regime in which both sectors approach comparable leakage. The exact-state parity-readout correction reduces but does not remove this replication signal ($p = 4.16 \times 10^{-8}$), indicating that the promoted sign and statistical conclusion are not an artefact of the mea-

Block	Scope	Circuits	QPU s	Fisher result
A+G	$n = 4$ replication plus readout	612	687	$\chi^2 = 140.671952$, $p = 3.77 \times 10^{-20}$
B-C $n = 6$	Early scaling, depths 4, 8, 14, 20	280	305	$\chi^2 = 46.531552$, $p = 1.88 \times 10^{-7}$
B-C $n = 8$	Early scaling, depths 4, 8, 14, 20	280	305	$\chi^2 = 29.420107$, $p = 2.68 \times 10^{-4}$

TABLE III. Compact Phase 2 hardware result table. A+G used jobs `ibm-run-7da8644af35021fb` and `ibm-run-6f9990bba1d90a12`; B-C used job `ibm-run-1f46ebd0da8912ff`. Raw-count SHA256 hashes are indexed in the publication package manifest.

sured readout-only parity flips.

The B-C block constrains the scaling claim. At $n = 8$, depths $4, 8, 14$ preserve the positive sign, with relative asymmetries $+3.33\,\%$, $+1.58\,\%$, and $+1.33\,\%$. At $n = 6$, two significant depths have the opposite sign, $-3.74\,\%$ at $d = 8$ and $-1.94\,\%$ at $d = 20$. The correct promoted conclusion is therefore narrow: strong $n = 4$ replication, same-day readout control, and mixed $n = 6, 8$ early scaling. Phase 2 falsifies a simple monotone scaling story and does not constitute broad quantum advantage, GUESS mitigation validation, or $n = 10, 12$ hardware validation.

A. Popcount-control follow-up

After the B-C scaling run, we executed the preregistered popcount control on `ibm_kingston`. The run used 360 parity-leakage circuits and five readout circuits. It tested five $n = 4$ initial states: the original even state $|0011\rangle$, an even same-popcount swap $|0101\rangle$, the original odd state $|0001\rangle$, an odd same-popcount swap $|0010\rangle$, and a higher-excitation odd state $|0111\rangle$. The jobs were `ibm-run-7d468e2b1e44b406` and `ibm-run-b3424c38cfe03c86`; the raw-count SHA256 hash is indexed in the publication package manifest.

The control is scientifically useful because it downgrades the causal interpretation. The original $|0011\rangle - |0001\rangle$ contrast is still significant in the middle-depth window ($+9.73\,\%$ at $d = 6$, $+17.13\,\%$ at $d = 8$, and $+14.42\,\%$ at $d = 10$), but it reverses by $d = 20$ ($-9.87\,\%$). Same-popcount within-sector swaps are also significant: the even swap has Fisher $p = 2.06 \times 10^{-19}$ across depths, and the odd swap has Fisher $p = 4.62 \times 10^{-14}$. Finally, the excitation-inversion arm shows that the popcount-3 odd state generally leaks more than the popcount-2 even state, with Fisher $p = 8.829457 \times 10^{-24}$ for the signed $|0011\rangle - |0111\rangle$ comparison. These results indicate that state preparation, layout, and excitation count materially contribute to the leakage response. The promoted interpretation is therefore a reproducible parity-sector/excitation-number correlated hardware phenomenon, not an isolated proof that DLA parity alone determines decoherence robustness. The same exact-state parity-readout correction preserves all four popcount-control comparison families: the corrected original contrast gives Fisher $p = 9.27 \times 10^{-18}$, the corrected within-even and within-odd swaps give $p = 4.47 \times 10^{-20}$ and $p = 3.59 \times 10^{-15}$, and the corrected excitation-inversion arm gives $p = 5.93 \times 10^{-18}$.

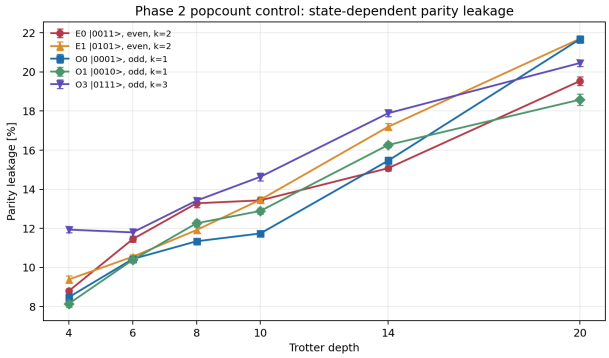


FIG. 6. Phase 2 popcount-control leakage curves. The original even–odd contrast remains visible in the middle-depth window, but same-popcount within-sector swaps and the popcount-3 odd state also show state-dependent leakage. The control supports a multifactorial parity-sector/excitation-number interpretation rather than a DLA-parity-only causal claim.

B. Phase 3 multi-device boundary

On 2026-05-06 we executed the next preregistered boundary test: a small second-backend repetition on `ibm_marrakesh`. The run used the same $n = 4$ parity-leakage observable at promoted depths $d \in \{4, 6, 8, 10, 14, 20\}$, plus a full-basis readout calibration on the same physical qubits [5,6,7,8]. The parity job `ibm-run-63e0a1af74a38c9c` submitted 148 circuits; the matching readout job `ibm-run-ddd29a2fbcaeed61` submitted all 16 computational-basis calibration states with 8192 shots per state. The calibration matrix is well conditioned (condition number 1.07570), with mean state retention 0.96999 and maximum measured parity-flip probability 0.04724.

The second-backend result does not reproduce the positive `ibm_kingston` sign. Table IV reports the raw relative asymmetry and the full-basis readout-corrected relative asymmetry. The only raw positive point, $d = 6$, becomes negative after readout correction, and all corrected promoted depths are non-positive.

This is a falsification of the strongest transfer claim. The Phase 1 and Phase 2 `ibm_kingston` signal remains a reproducible same-device raw-count phenomenon, and

d	Raw $A(d)$	Full-basis corrected $A(d)$
4	−2.53 %	−31.60 %
6	+3.69 %	−9.62 %
8	−0.13 %	−3.58 %
10	−2.48 %	−0.20 %
14	−7.43 %	−4.56 %
20	−3.46 %	−5.00 %

TABLE IV. Phase 3 second-backend boundary on `ibm_marrakesh`. The full-basis readout correction uses the matching 16-state calibration job on physical qubits [5,6,7,8]. The promoted conclusion is backend sensitivity, not backend-stable parity protection.

the popcount-control structure remains scientifically informative, but the sign and magnitude are not backend-stable under this `ibm_marrakesh` layout and calibration window. Any future positive universality claim therefore requires layout-randomised or multi-backend statistics rather than a single-device replication.

This point is central to the paper rather than a caveat. The data do not support the strong statement that DLA parity alone determines leakage robustness. They support a narrower and more useful statement: parity sector, excitation number, prepared computational-basis state, layout, backend, and calibration window jointly shape the measured leakage response. The `ibm_kingston` observation remains real, but the progressively stronger controls deliberately demote the causal claim from "DLA-parity protection" to "parity-sector/excitation-number correlated hardware response with backend and layout sensitivity."

At the time of execution, authenticated live QPU access for this campaign was limited to IBM Quantum hardware. The repository contains provider-neutral execution and artefact-packaging surfaces for other gate-model, trapped-ion, neutral-atom, photonic, annealing, analogue, and simulator targets, but those software routes are not evidence of live non-IBM execution. The preregistered parity-sector, readout-mitigation, and layout-randomisation payloads are intended to be rerun on additional hardware families when compute allocations and provider access become available. Until then, the result is an IBM-device and IBM-backend sensitivity statement, not a cross-provider universality claim.

C. Phase 3 state/layout randomisation

The second Phase 3 control asked whether the original-state contrast survives controlled changes in both physical layout and same-sector computational-basis state. The campaign submitted 480 parity circuits to `ibm_marrakesh` (job `ibm-run-aabcf620230b1438`) plus a layout-matched five-state readout calibration (`ibm-run-eea172711aa52b78`). It used depths $d \in \{6, 8, 10, 14\}$, three connected four-qubit layouts, eight repetitions per state/depth/layout cell, and live transpilation gates that accepted a maximum observed depth of 294 under a 700-depth cap and 762 total gates under a 1500-gate cap.

The preregistered analysis again rejects a simple symmetry-only interpretation. For the original even–odd comparison (`E0-O0`), the mean leakage difference is positive but small (+0.006312), with six positive and six negative layout-depth cells. Within-sector controls are not null: the even-sector state swap (`E0-E1`) has Fisher $p = 2.52 \times 10^{-59}$ and the odd-sector state swap (`O0-O1`) has Fisher $p = 1.40 \times 10^{-27}$. The excitation-inversion arm (`E0-O3`) has seven negative and five positive cells. The derived decision flags record that the layout spread exceeds the mean original contrast, the original contrast has mixed sign, and both within-sector controls are significant.

This does not erase the Phase 1/2 `ibm_kingston` observation. It sharpens the mechanism boundary: parity sector, excitation count, prepared computational-basis state, physical layout, backend, and calibration window are all experimentally relevant at this scale. Consequently, this paper reports a reproducible same-device leakage asymmetry and a sequence of controls that falsify progressively stronger causal claims, not a layout-independent DLA-protection mechanism.

VI. CONCLUSION

We have measured a parity-sector and excitation-number correlated decoherence asymmetry of the XY Hamiltonian's dynamical Lie algebra on real superconducting hardware and reproduced the $n = 4$ signal in an independent same-device raw-count continuation. The Phase 1 observation gives a (10.8 ± 1.1) % mean asymmetry across depths $d \geq 4$ and agrees quantitatively with an independent classical simulator prediction. The Phase 2 A+G block independently strengthens the $n = 4$ raw-count observation with Fisher $p = 3.77 \times 10^{-20}$, while the B-C block shows that scaling is mixed rather than monotone. The popcount-control follow-up shows that excitation count and state choice materially contribute to the signal. The Phase 3 `ibm_marrakesh` boundary test then shows that the positive `ibm_kingston` sign is not backend-stable after matching full-basis readout correction. The state/layout randomisation control then shows that layout and same-sector state choice can be comparable to, or larger than, the mean original contrast. The validated claim is therefore an $n = 4$ same-device raw-count hardware phenomenon with constrained early scaling evidence, backend and layout sensitivity, and an explicitly limited causal interpretation, not a broad quantum-advantage result and not a DLA-parity-only result. Existing data also support two no-new-QPU follow-up conclusions: exact-state parity-readout correction preserves the main Phase 2 signs, and parity survival is smooth enough versus depth to motivate a future

GUESS-style folded-noise protocol. The latter is only a readiness check, not a demonstrated zero-noise extrapolation.

DATA AND CODE AVAILABILITY

All raw count dictionaries, per-circuit metadata, analysis scripts, and figure-generation code are open-sourced at anulum/scpn-quantum-control under the AGPL-3.0 licence. The Phase 1 dataset lives in `data/phase1_dla_parity/` (four JSON files, 342 circuits at $n = 4$), and the full analysis can be reproduced with

```
python scripts/analyse_phase1_dla_parity.py
```

The Phase 2 A+G raw-count dataset lives in `data/phase2_dla_parity/`. The Phase 2 B-C raw-count dataset lives in `data/phase2_scaling_bc/`. The exact Phase 2 reproduction commands, job identifiers, figure artefacts, and SHA256 hashes are indexed in `docs/publication/publication_phase2_package_2026-05-05.md`. The popcount-control dataset lives in `data/phase2_popcount_control/` and is reproduced with `scripts/analyse_phase2_popcount_control.py`. The no-new-QPU readout and GUESS-readiness cross-checks are reproduced with `scripts/analyse_phase2_readout_mitigation.py` and `scripts/analyse_phase2_guess_calibration.py`; their derived JSON summaries live in `data/phase2_readout_mitigation/` and `data/phase2_guess_calibration/`. The Phase 3 second-backend `ibm_marrakesh` raw counts, row metrics, and full-basis readout-corrected summaries live in `data/phase3_multidevice_dla/` and are indexed in `docs/campaigns/phase3_multidevice_dla_manifest_2026-05-06.md`. The matching full-basis readout calibration lives in `data/readout_full_basis/` and is indexed in `docs/campaigns/readout_full_basis_manifest_2026-05-06.md`. The submission drivers are `scripts/phase3_multidevice_dla_ibm.py` and `scripts/readout_full_basis_calibration_ibm.py`. The Phase 3 state/layout-randomisation raw counts and generated row, layout, and readout metrics live in `data/phase3_state_layout_dla/`; they are reproduced with `scripts/analyse_phase3_state_layout_dla.py` and indexed in `docs/campaigns/phase3_state_layout_dla_manifest_2026-05-07.md`. The state/layout submission driver is `scripts/phase3_state_layout_dla_ibm.py`. The campaign archive carries Zenodo DOI 10.5281/zenodo.18821929.

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